Practice Test — Chapter 8: Energy

Year 10 Science (Victorian Curriculum, VCSSU132)

Name: ____________________ **Date:** ____________

Time allowed: 50 minutes **Total marks:** /50

Instructions: Answer all questions. Show full working for calculations and include units. Use g = 9.8 m/s². The answer key is at the end — try the whole test before checking.

Useful formulas: $W = Fs$ | $K = \frac{1}{2}mv^2$ | $U_g = mgh$ | Efficiency = (useful energy out ÷ energy in) × 100%

Section A — Multiple choice (10 marks, 1 mark each)

Circle the best answer.

1. Energy is measured in: - A. newtons - B. joules - C. watts - D. metres

2. Kinetic energy is the energy an object has due to its: - A. position - B. shape - C. motion - D. temperature

3. Which of the following is a form of **potential** energy? - A. thermal energy - B. sound energy - C. gravitational energy - D. light energy

4. Temperature is a measure of the **average kinetic energy** of the particles, whereas thermal energy is the: - A. average potential energy of the particles - B. total kinetic energy of all the particles - C. heat lost to the surroundings - D. mass of the object

5. Heat transfer through a vacuum (such as from the Sun to Earth) occurs by: - A. conduction - B. convection - C. radiation - D. insulation

6. Heat transfer by the movement of particles in a fluid is called: - A. conduction - B. convection - C. radiation - D. reflection

7. The law of conservation of energy states that energy: - A. is always lost over time - B. can be created but not destroyed - C. cannot be created or destroyed, only transformed or transferred - D. always increases in a system

8. No energy conversion is 100% efficient because some energy is always: - A. created - B. destroyed - C. lost from the system (usually as heat/sound) - D. converted to mass

9. A material that is a **poor** conductor of heat is called a(n): - A. conductor - B. insulator - C. radiator - D. metalloid

10. Crumple zones reduce injury in a car crash by: - A. increasing the force on the occupants - B. increasing the time and distance of the collision, reducing the force - C. making the car heavier - D. storing the kinetic energy as light

---

Section B — Short answer (14 marks)

11. Define **energy** and define **work**. (2 marks)

12. Explain the difference between **temperature** and **heat**. (2 marks)

13. Name the **three** methods of heat transfer and give a one-line description of each. (3 marks)

14. State the **law of conservation of energy.** (1 mark)

15. Explain why an object with a **lower temperature** can have **more thermal energy** than an object at a higher temperature. (2 marks)

16. Two blocks are in **thermal equilibrium**. Explain what this means in terms of temperature and heat transfer. (2 marks)

17. Explain why metals are good conductors of heat, using the particle model. (2 marks)

Section C — Calculations (16 marks)

Show all working and include units.

18. A weightlifter applies a force of 80 N to lift a barbell 1.5 m off the ground. Calculate the **work** done. (2 marks)

19. A 2 kg ball is moving at 6 m/s. Calculate its **kinetic energy.** (3 marks)

20. A 60 kg diver stands on a diving board 10 m above the water. Calculate the diver's **gravitational potential energy.** (3 marks)

21. A light globe is supplied with 75 J of electrical energy and produces 15 J of light energy. (4 marks)

a) Calculate the **efficiency** of the light globe. (2 marks)

b) How much energy is "wasted", and what form does it usually take? (2 marks)

22. An electric stove takes in 1500 J of electrical energy and produces 200 J of light and 1300 J of thermal energy. (4 marks)

a) Calculate the efficiency if the **useful** output is the thermal (heat) energy. (2 marks)

b) Sketch a labelled **Sankey diagram** for this stove. (2 marks)

Section D — Extended response (10 marks)

23. A rubber ball is dropped from a height onto the ground. It bounces several times, each time reaching a lower height, until it stops. (5 marks)

a) Describe the energy transformations as the ball falls, hits the ground, and bounces back up. (3 marks)

b) Explain why the ball does not bounce back to its original height, using the idea of energy efficiency. (2 marks)

24. Describe how a refrigerator keeps food cold. In your answer, refer to **heat transfer, the coolant, work, and the law of conservation of energy.** (5 marks)

—

ANSWER KEY — Energy

Section A 1. B — joules 2. C — motion 3. C — gravitational energy 4. B — total kinetic energy of all the particles 5. C — radiation 6. B — convection 7. C — cannot be created or destroyed, only transformed or transferred 8. C — lost from the system (usually as heat/sound) 9. B — insulator 10. B — increasing the time and distance of the collision, reducing the force

Section B

11. Energy = the ability to do work (the capacity to cause change). Work = the energy transferred when an applied force moves an object some distance. (2)

12. Temperature = the average kinetic energy of the particles in an object. Heat = thermal energy transferred between objects due to a temperature difference. (2)

13. Conduction = transfer through a substance via particle collisions; Convection = transfer by the movement of particles in a fluid (currents); Radiation = transfer of thermal energy without particles (e.g. infrared/EMR). (3)

14. Energy cannot be created or destroyed; it can only be transformed into other forms or transferred to other objects. (1)

15. Thermal energy depends on the **total** kinetic energy of **all** the particles, so an object with more particles (more mass) can have more total thermal energy even if its particles move slower (lower temperature). (2)

16. They are at the **same temperature**, so there is **no net transfer** of thermal energy between them. (2)

17. Metals have particles packed closely together, so vibrating particles collide frequently and pass on kinetic energy quickly; (free electrons also help carry energy). (2)

Section C

18. $W = Fs = 80 \times 1.5 =$ **120 J** (2)

19. $K = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 6^2 = \frac{1}{2} \times 2 \times 36 =$ **36 J** (3)

20. $U_g = mgh = 60 \times 9.8 \times 10 =$ **5880 J (5.88 kJ)** (3)

21. a) Efficiency = $(15 \div 75) \times 100 =$ **20%** (2) b) Wasted energy = $75 - 15 =$ **60 J**, usually as **thermal (heat) energy.** (2)

22. a) Efficiency = $(1300 \div 1500) \times 100 =$ **86.7%** (2) b) Sankey diagram: a wide arrow for 1500 J input splitting into a thick arrow of 1300 J thermal (useful) and a thin arrow of 200 J light (wasted). Arrow widths proportional to energy. (2)

Section D

23. a) (3) At the top the ball has maximum gravitational potential energy (GPE). As it falls, GPE converts to kinetic energy (KE), reaching maximum KE just before impact. On hitting the ground, KE converts to elastic potential energy (the ball compresses), then back to KE as it bounces up, converting again to GPE. b) (2) Energy is lost from the system with each bounce — as heat (friction/air resistance) and sound — so less energy is available to convert back to GPE; the system is not 100% efficient, so each bounce is lower until the ball stops. (Energy is not destroyed, just transformed/transferred.)

24. Marking points (5): Heat naturally moves from hot to cold, so moving heat **out** of cold food requires **work** (energy). **Electrical energy** powers a compressor/pump that circulates a **coolant (refrigerant)**. The coolant **absorbs thermal energy** from inside the fridge (evaporating to a gas), then is compressed and **releases that heat outside** (condensing to a liquid) before the cycle repeats. The **law of conservation of energy** holds — energy is transferred/transformed, not created or destroyed (wasted energy is heat and sound). *(Award 1 mark each, max 5.)*

END OF TEST